

Differentiation and Integration Formulas

Odisha CHSE Class 11 & 12 Science

1 Differentiation Formulas

Basic Rules

$$\begin{aligned}\frac{d}{dx}(c) &= 0 \\ \frac{d}{dx}(x^n) &= nx^{n-1} \\ \frac{d}{dx}(\sqrt{x}) &= \frac{1}{2\sqrt{x}} \\ \frac{d}{dx}(x^{-n}) &= -nx^{-n-1}\end{aligned}$$

Trigonometric Functions

$$\begin{aligned}\frac{d}{dx}(\sin x) &= \cos x \\ \frac{d}{dx}(\cos x) &= -\sin x \\ \frac{d}{dx}(\tan x) &= \sec^2 x \\ \frac{d}{dx}(\cot x) &= -\csc^2 x \\ \frac{d}{dx}(\sec x) &= \sec x \tan x \\ \frac{d}{dx}(\csc x) &= -\csc x \cot x\end{aligned}$$

Exponential and Logarithmic

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(a^x) = a^x \ln a$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}(\log_a x) = \frac{1}{x \ln a}$$

Inverse Trigonometric

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$$

$$\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\csc^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}$$

Rules

$$\text{Product Rule: } \frac{d}{dx}[u \cdot v] = u'v + uv'$$

$$\text{Quotient Rule: } \frac{d}{dx} \left[\frac{u}{v} \right] = \frac{u'v - uv'}{v^2}$$

$$\text{Chain Rule: } \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

2 Integration Formulas

Basic Integrals

$$\begin{aligned}\int c \, dx &= cx + C \\ \int x^n \, dx &= \frac{x^{n+1}}{n+1} + C \quad (n \neq -1) \\ \int \frac{1}{x} \, dx &= \ln |x| + C \\ \int e^x \, dx &= e^x + C \\ \int a^x \, dx &= \frac{a^x}{\ln a} + C\end{aligned}$$

Trigonometric Functions

$$\begin{aligned}\int \sin x \, dx &= -\cos x + C \\ \int \cos x \, dx &= \sin x + C \\ \int \sec^2 x \, dx &= \tan x + C \\ \int \csc^2 x \, dx &= -\cot x + C \\ \int \sec x \tan x \, dx &= \sec x + C \\ \int \csc x \cot x \, dx &= -\csc x + C\end{aligned}$$

Inverse Trigonometric

$$\begin{aligned}\int \frac{1}{\sqrt{1-x^2}} \, dx &= \sin^{-1} x + C \\ \int \frac{1}{1+x^2} \, dx &= \tan^{-1} x + C \\ \int \frac{1}{x\sqrt{x^2-1}} \, dx &= \sec^{-1} x + C\end{aligned}$$

3 Integration Techniques

Substitution

$$\int f(g(x)) \cdot g'(x) dx = \int f(u) du$$

Integration by Parts

$$\int u dv = uv - \int v du$$

Partial Fractions

$$\begin{aligned}\frac{1}{(x-a)(x-b)} &= \frac{A}{x-a} + \frac{B}{x-b} \\ \frac{1}{(x-a)^2} &= \frac{A}{x-a} + \frac{B}{(x-a)^2} \\ \frac{1}{x^2+bx+c} &= \frac{Ax+B}{x^2+bx+c}\end{aligned}$$

4 Examples

Differentiation

1. Basic Rule:

$$\frac{d}{dx}(x^5) = 5x^4, \quad \frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$$

2. Trigonometric:

$$\frac{d}{dx}(\sin x) = \cos x, \quad \frac{d}{dx}(\tan x) = \sec^2 x$$

3. Exponential and Logarithmic:

$$\frac{d}{dx}(e^x) = e^x, \quad \frac{d}{dx}(\ln x) = \frac{1}{x}$$

4. Inverse Trigonometric:

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, \quad \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

5. Product Rule:

$$\frac{d}{dx}(x \cdot \sin x) = x \cos x + \sin x$$

6. Quotient Rule:

$$\frac{d}{dx} \left(\frac{x^2}{\sin x} \right) = \frac{2x \sin x - x^2 \cos x}{\sin^2 x}$$

Integration

1. Basic Integral:

$$\int x^3 dx = \frac{x^4}{4} + C, \quad \int \frac{1}{x} dx = \ln |x| + C$$

2. Trigonometric:

$$\int \cos x dx = \sin x + C, \quad \int \sec^2 x dx = \tan x + C$$

3. Exponential and Logarithmic:

$$\int e^x dx = e^x + C, \quad \int a^x dx = \frac{a^x}{\ln a} + C$$

4. Inverse Trigonometric:

$$\int \frac{1}{1+x^2} dx = \tan^{-1} x + C, \quad \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + C$$

5. Substitution: Let $u = 1 + x^2$, then $du = 2x dx$.

$$\int \frac{2x}{1+x^2} dx = \int \frac{du}{u} = \ln |u| + C = \ln(1+x^2) + C$$

6. Integration by Parts: Let $u = x$, $dv = e^x dx$. Then $du = dx$, $v = e^x$.

$$\int x e^x dx = x e^x - \int e^x dx = x e^x - e^x + C$$